Sahilpreet Singh Sidhu

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TECHNICAL SKILLS

AI and ML: LLMs, RAG, LLM agents (tool calling, MCP), fine-tuning (LoRA, T5), reinforcement learning (PPO, Unity ML-Agents), PyTorch, Hugging Face Transformers, embeddings, FAISS, Ollama, ONNX, evaluation harnesses
Languages: Python, C++, TypeScript, JavaScript, C#, SQL
Backend and Web: FastAPI, React, REST, WebSockets, SQLite
Infra: Linux, Docker, systemd, CUDA, Cloudflare (Pages, Tunnels)

PROJECTS

AI · LLMS · RAG

RaceVoice: AI race commentator | *Python, PyTorch, Hugging Face Transformers* 2026

- Built a single sub-0.5B model that turns race telemetry into spoken commentary: words and voice from one network (a decoder predicting text and audio codes), not an LLM feeding a separate TTS.
- Pruned the vocabulary from 156,802 to 9,443 tokens, cutting the model from 499M to 366M parameters and training memory about 7x; energy tokens select a shouted or calm delivery at inference.
- ▸ **Impact:** voices a fully automated 24/7 YouTube racing channel, at 50 tok/s with first audio in 0.57 s, on the same 8 GB laptop that runs the game, OBS and the stream.

AI Racing Drivers: ten RL personalities | *Python, Unity ML-Agents (PPO), C#, ONNX* 2026

- Trained ten driver personalities with PPO on live simulation from raycast and telemetry inputs, up to 120 cars in parallel: one 11M-step base policy, then per-personality reward weights for overtaking, defending, racing line and braking.
- The base policy completes timed laps on all 10 circuits from a 16-car standing grid at about 223 km/h average.
- ▸ **Impact:** every car on the automated YouTube racing channel is one of these models; races run AI against AI with no human input.

2DVideoGen: smallest model for anime-style animation | *Python, PyTorch, Transformers (T5)* 2026
github.com/SahilSidhu7/2DVideoGen

- Trained a 60M model to turn plain-English prompts into a scene-script DSL, rendered to video and scored by an automatic critic; 100% of held-out prompts parse and render, trained in under 9 minutes on CPU.
- Ran controlled parameter-vs-data experiments: three data revisions at a fixed 60M cut the generalisation gap from 65.7 to 15 points, while 3.7x the parameters (220M) on identical data raised every metric but not the rendered video.
- ▸ **Impact:** multi-character animation from a model small enough to train on a laptop, and evidence that data design, not model size, was the bottleneck; all 24 attempts logged with their real causes.

PYTHON · REACT · FASTAPI · LINUX · DOCKER

miniVE: disposable Linux environments | *Docker, Linux, Tauri, Svelte, TypeScript* 2026
github.com/SahilSidhu7/miniVE

- Desktop app that spins up isolated Python, Node or Ubuntu environments backed by Docker containers in seconds, with runtime versions fetched live from Docker Hub, multi-tab terminals and in-app port preview.
- ▸ **Impact:** a throwaway Linux box for testing repos or sandboxing AI coding agents without breaking the host machine; deleting one removes the container, volume and every trace. Ships v0.5 for Windows, macOS and Linux.

ServManager: server dashboard and phone remote | *Python, FastAPI, React, WebSockets, systemd* 2026
github.com/SahilSidhu7/servmanager

- FastAPI and React app with live CPU, memory and disk over WebSocket, a browser script editor with scheduled runs and streamed output, xterm.js SSH terminals, and a PIN-locked phone remote.
- ▸ **Impact:** routine server work (restarting services, running scripts, checking health) done from a phone; one port, no database, installs as a systemd service from a .deb, and runs my own home server.

EDUCATION

Chitkara University, B.Tech in Computer Science 2024 to 2028 (expected)
CGPA: 9.27 / 10 | Coursework: Data Structures and Algorithms, Operating Systems, Computer Networks, OOP

CERTIFICATIONS

Microsoft Azure AI Fundamentals (AI-900), Azure Fundamentals (AZ-900), Azure Data Fundamentals (DP-900); Full Stack Open, University of Helsinki (7 ECTS, grade 5/5); Generative AI for Everyone, DeepLearning.AI